

Welfare and Spending Effects of Consumption Stimulus Policies

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- ▶ **This paper:** Aim to evaluate three consumption stimulus policies in a HA model consistent with data on liquid wealth and *intertemporal* MPCs

Related literature

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- ▶ **High MPCs and impatience:** Parker (2017)

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- ▶ Robustness Exercise: HANK model

Quantitative Micro Realism

- ▶ Idiosyncratic income process: Friedman/Muth (transitory and permanent shocks)

$\mathbf{p}$ — ‘permanent income’

ξ — ‘transitory income shock’

ψ — ‘permanent income shock’

$$\mathbf{p}_{t+1} = \mathcal{G}^e \mathbf{p}_t \psi_{t+1}$$

$$y_{t+1} = \mathbf{p}_{t+1} \xi_{t+1}$$

- ▶ $\mathcal{G}^e$: education-specific income growth
- ▶ Evidence for permanent shocks: See Crawley, Holm, and Tretvoll (2024)

Preferences, Beliefs, and Wealth

Infinite horizon model: target wealth depends on 'Growth Impatience' condition:

$$\underbrace{\left(\frac{(R \beta^{e,i})^{1/\gamma}}{\mathcal{G}^e \mathbb{E}[\psi^{-1}]} \right)}_{\text{'Growth Patience Factor'}} < 1 \quad (1)$$

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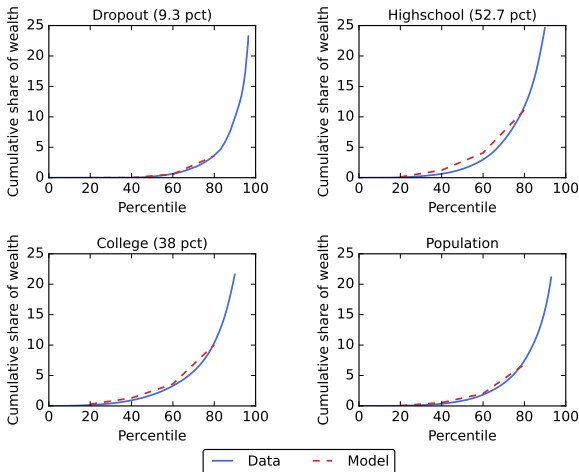
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Consistency With Micro Evidence (1)

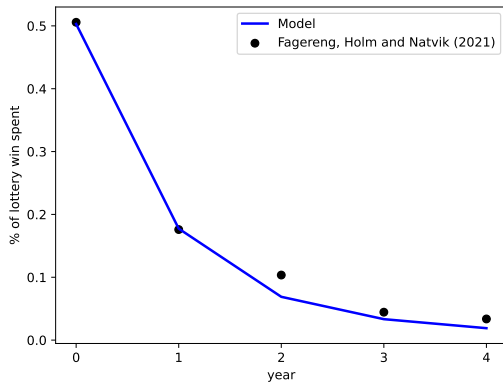
Liquid Wealth from
Survey of Consumer
Finances (SCF)



- ▶ Education groups: $e \in \{\text{'Dropout'}, \text{'Highschool' and 'College'}\}$
- ▶ Each group has distribution of discount factors $\beta_{e,i}$

Consistency With Micro Evidence (2)

Intertemporal MPC from Fagereng, Holm, Natvik (2021)



Modeling device: 'Splurge' in consumption

Splurge consumption

- ▶ Exogenous fraction of income directly consumed
- ▶ Model consistent with spending patterns over time after a transitory income shock
- ▶ Evidence: High liquid wealth hh also have high MPCs
 - ▶ Kueng (2018); Crawley and Kuchler (2023); Graham and McDowall (2024)
- ▶ Possible microfoundations:
 - ▶ Spending on durables (Browning and Crossley, 2009; Laibson et al., 2022)
 - ▶ A form of present bias (Indarte et al., 2024, Maxted et al., 2024)
- ▶ Robustness: Model w/o splurge consumption

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 - ▶ Welfare (only recession-related welfare impact)

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 - ▶ Very similar pattern for cumulative multipliers

Model

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- ▶ Remaining consumption $\mathbf{c}_{opt,i,t}$ is chosen to maximize utility

$$\sum_{t=0}^{\infty} \beta_{e,i}^t (1 - D)^t \mathbb{E}_0 u(\mathbf{c}_{opt,i,t}). \quad (2)$$

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- ▶ Budget constraint, given existing market resources $\mathbf{m}_{i,t}$ and income state, and a no-borrowing constraint:

$$\mathbf{m}_{i,t+1} = R \underbrace{(\mathbf{m}_{i,t} - \mathbf{c}_{sp,i,t} - \mathbf{c}_{opt,i,t})}_{\geq 0 \text{ (no-borrowing constraint)}} + \mathbf{y}_{i,t+1} \quad (3)$$

(R : exogenous gross interest rate)

Income process

- Income subject to transitory, unempl. and permanent shocks

$$\mathbf{y}_{i,t} = \begin{cases} \xi_{i,t}\mathbf{p}_{i,t}, & \text{if employed} \\ 0.7\mathbf{p}_{i,t}, & \text{if unemployed for } \leq 2q \\ 0.5\mathbf{p}_{i,t}, & \text{if unemployed } \geq 2q \end{cases} \quad (4)$$

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- ▶ Model is a simplified model of households (no heterogeneity in hh size)
- ▶ Replacement rates reflect some degree of hh insurance (Rothstein and Valetta, 2017)

Employment status and recessions

- ▶ Employment status is subject to a Markov process
 - ▶ Employed consumer: continue being employed or become unemployed
 - ▶ Unemployed consumers: receives benefits for two quarters
- ▶ Bureau of Labor Statistics: Report unemployment rates by education group
- ▶ Recession is given by an MIT shock
 - ▶ Unemployment rate doubles in each education group
 - ▶ Expected length of unemployment increases from 2 to 4q
 - ▶ End of recession occurs as a Bernoulli process calibrated for an avg. rec. length of 6q

Aggregate demand effects

(as in Krueger, Mitman and Perri, 2016)

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$$AD(C_t) = \begin{cases} \left(\frac{C_t}{\tilde{C}}\right)^\kappa, & \text{if in a recession} \\ 1, & \text{otherwise,} \end{cases} \quad (5)$$

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- ▶ Idiosyncratic income in the extension model is then given by

$$\mathbf{y}_{AD,i,t} = AD(C_t)\mathbf{y}_{i,t}. \quad (6)$$

Parameters — by education group

[More parameters](#)[Policy parameters](#)

Parameters calibrated for each education group			
	Dropout	Highschool	College
Percent of population	9.3	52.7	38.0
Avg. quarterly PI of “newborn” agent (\$1000)	6.2	11.1	14.5
Std. dev. of log(PI) of “newborn” agent	0.32	0.42	0.53
Avg. quarterly gross growth rate of PI (Γ_e)	1.0036	1.0045	1.0049
Unemployment rate in normal times (percent)	8.5	4.4	2.7
Probability of entering unemployment (π_{eu}^e , percent)	6.2	3.1	1.8
Probability of leaving unemployment (π_{ue})	0.667	0.667	0.667

- Mincer (1991) and Elsby and Hobjin (2010): Education groups differ in the incidence of unemployment, not its duration

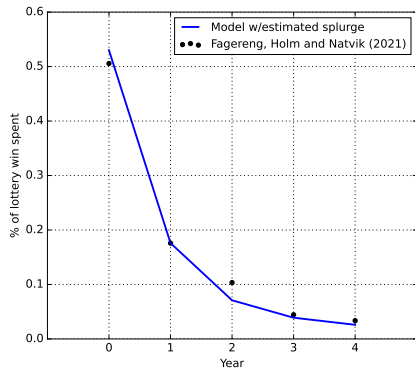
Results

Untargeted moments (1)

Non-targeted moments by wealth quartile

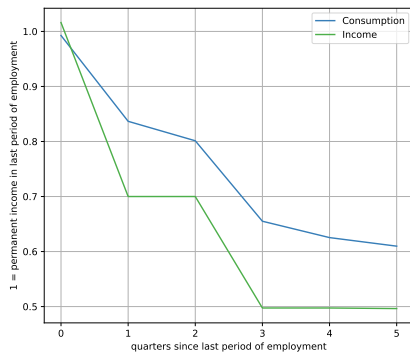
	WQ 4	WQ 3	WQ 2	WQ 1
Percent of liquid wealth (data)	0.14	1.60	8.51	89.76
Percent of liquid wealth (model, baseline)	0.12	0.98	3.85	95.0
Percent of liquid wealth (model, Splurge=0)	0.10	1.07	4.24	94.60
Avg. lottery-win-year MPC (model, incl. splurge)	0.74	0.61	0.48	0.32
Avg. lottery-win-year MPC (model, splurge=0)	0.69	0.53	0.36	0.14

Untargeted moments (2)



Share of lottery win spent

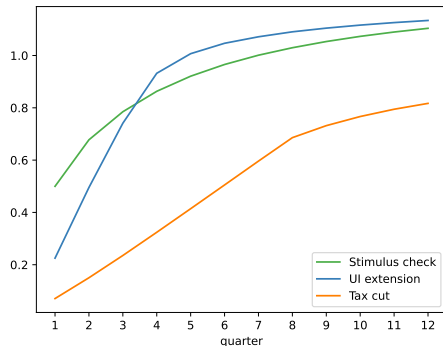
- ▶ Ganong and Noel (2019): UI expiry $\Rightarrow$ drop of 12 percent (month)
- ▶ Our model $\Rightarrow$ drop of 18 percent (quarter)



Spending upon expiry of UI benefits

Multipliers

$$M_t^P = \frac{\text{NPV of induced consumption up to } t}{\text{NPV of the cost of the policy}}$$

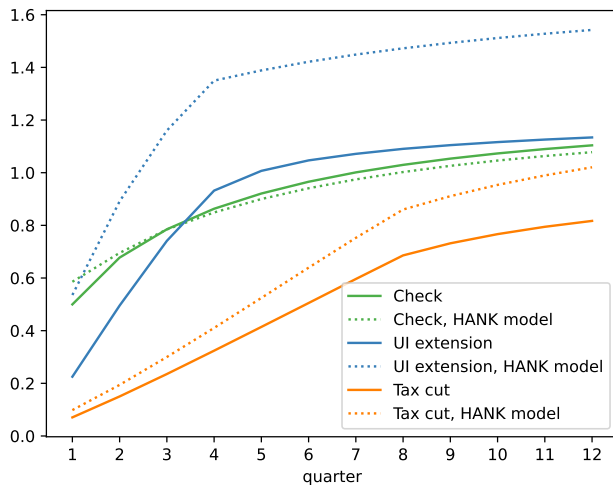


	Stimulus check	UI extension	Tax cut
10y-horizon Multiplier (no AD effect)	0.88	0.91	0.85
10y-horizon Multiplier (AD effect)	1.23	1.21	0.98
Share of policy expenditure during recession	100.0%	79.6%	57.8 %

Robustness: Multipliers in a HANK and SAM model — Setup

- ▶ Evaluate the policies in a relatively standard HANK and SAM model (Du, 2024)
- ▶ New Keynesian: Monopolistic competition + sticky prices
- ▶ Search and matching: Random search, labor market tightness affects job finding and vacancy filling probabilities
- ▶ Government policy: Monetary and fiscal rules
- ▶ Fiscal multipliers through an intertemporal Keynesian cross mechanism
However: No state dependence
- ▶ Solution method $\Rightarrow$ cannot evaluate effects starting in a deep recessionary state
This also implies that we cannot use our welfare measure

Robustness: Multipliers in a HANK and SAM model — Results



Welfare measure

- Aim: Welfare measure does not reflect benefits of redistribution in “normal” times

$$\mathcal{W}(\text{policy}, \text{Rec}, AD) = \frac{1}{\mathcal{N}} \sum_{i=1}^N \sum_{t=0}^{\infty} \frac{1}{R^t} \frac{u(\mathbf{c}_{it, \text{policy}, \text{Rec}, AD}) - u(\mathbf{c}_{it, \text{none}, \text{Rec}, AD})}{u'(\mathbf{c}_{it, \text{normal}})}$$

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where $\mathcal{N} = NPV(\text{policy}, Rec, AD)$

- ▶ Normal times: $\mathcal{W}(\text{policy}, 0, 0) = 1$ (for $\Delta \mathbf{c}_{it} \approx 0$)

Welfare results

	Stimulus check	UI extension	Tax cut
$\mathcal{W}(\text{policy}, Rec = 0, AD = 0)$	0.96	0.85	0.99
$\mathcal{W}(\text{policy}, Rec = 1, AD = 0)$	1.00	1.83	0.97
$\mathcal{W}(\text{policy}, Rec = 1, AD = 1)$	1.35	2.15	1.11

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- ▶ $AD = 1$: Stimulating spending during recession increases measure for all policies

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 - ▶ is more easily scaled up to provide more stimulus
- ▶ The tax cut is both poorly targeted and may yield substantial spending after the recession is over
- ▶ Framework can be used to evaluate other candidate policies

Thank you for your attention!

- ▶ Access the paper, presentation slides and code at:
<https://github.com/econ-ark/HAFiscal>



Appendix

Parameters — same for all types

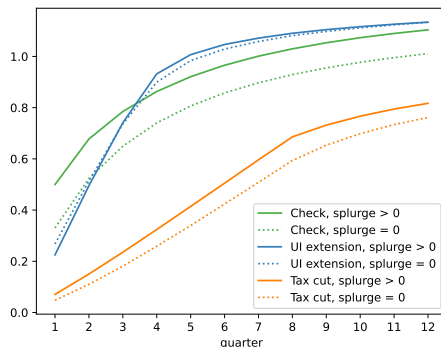
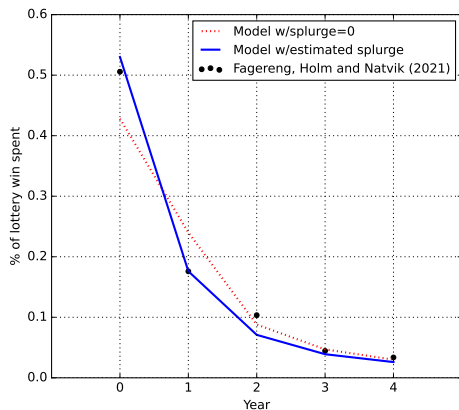
Parameter	Notation	Value
Risk aversion	γ	2.0
Splurge	ς	0.249
Survival probability, quarterly	$1 - D$	0.994
Risk free interest rate, quarterly (gross)	R	1.01
Standard deviation of transitory shock	σ_{ξ}	0.346
Standard deviation of permanent shock	σ_{ψ}	0.0548
Unemployment benefits replacement rate (share of PI)	ρ_b	0.7
Unemployment income w/o benefits (share of PI)	ρ_{nb}	0.5
Avg. duration of unemp. benefits in normal times (quarters)		2
Avg. duration of unemp. spell in normal times (quarters)		1.5
Consumption elasticity of aggregate demand effect	κ	0.3

Go back

Parameters describing the policies

Parameters describing policy experiments	
Parameter	Value
Change in unemployment rates in a recession	$\times 2$
Expected unemployment spell in a recession	4 quarters
Average length of recession	6 quarters
Size of stimulus check	\$1,200
PI threshold for reducing check size	\$100,000
PI threshold for not receiving check	\$150,000
Extended unemployment benefits	4 quarters
Length of payroll tax cut	8 quarters
Income increase from payroll tax cut	2 percent
Belief (probability) that tax cut is extended	50 percent

Robustness: Model w/o splurge consumption



	Stimulus check	UI extension	Tax cut
$\mathcal{W}(\text{policy}, Rec = 1, AD = 1)$	1.27(1.35)	2.12(2.15)	1.09(1.11)